

BTS-V(15)-11.12-1552

Reg. No.

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B.Tech. Degree V Semester Examination November 2017**ME 15-1506 POWER PLANT ENGINEERING**
(2015 Scheme)

Time : 3 Hours

Maximum Marks : 60

PART A
(Answer ALL questions)

(10 × 2 = 20)

- I. (a) Define load factor and demand factor.
- (b) Which types of power plants are used as base load power plants? Why are such power plants used as base load power plants?
- (c) What is a Surge tank? What is its function?
- (d) Explain the thermodynamic cycle of a simple gas turbine using T-s diagram.
- (e) Why diesel engines are heavier than petrol engines for the same power output? What is the range of compression ratios of diesel engines?
- (f) What are the advantages of gas turbines over diesel engines when used in power plants?
- (g) List any four types of Nuclear reactors.
- (h) What are the factors that determine the selection of methods for coal burning?
- (i) What is solar constant? What is its average value? What is solar insolation?
- (j) Explain the principle of ocean thermal energy conversion system.

PART B

(4 × 10 = 40)

- II. (a) A power station has to meet the following load demand: (5)
Load A-50 kW between 10 am and 6 pm. Load B-30 kW between 6 pm and 10 pm. Load C-20 kW between 4 pm and 10 am.
Plot the daily load curve and determine (i) diversity factor (ii) units generated per day (iii) load factor.
- (b) At the end of a power distribution system, a certain feeder supplies three distribution transformers, each one supplying a group of customers whose connected loads are as under: (5)

Transformer	Load	Demand factor	Diversity of groups
Transformer No.1	12 kW	0.65	1.5
Transformer No.2	14 kW	0.6	3.5
Transformer No.3	16 kW	0.7	1.5

If the diversity factor among the transformers is 1.35, find the maximum load on the feeder.

OR

- III. (a) The annual peak load on a 30 MW power station is 25 MW. The power station supplies load having maximum demands of 10 MW, 8.5 MW, 5 MW and 4.5 MW. The annual load factor is 0.45. Find. (5)
- (i) Average load.
(ii) Diversity factor.
(iii) Demand factor.
- (b) In a steam power plant the capital cost of power generation equipment is ₹2500000. The useful life of the plant is 25 years and salvage value of the plant is ₹100000. Determine by sinking fund method the amount to be saved annually for replacement if the rate of annual compound interest is 5%. (5)
- IV. (a) What are the starting methods used in large diesel engine power plants? Explain any one method. (5)
- (b) Draw the layout of a gas turbine power plant and explain its working. (5)
- OR
- V. (a) Derive the equation for the thermal efficiency of a simple diesel engine cycle. (5)
- (b) What are the methods used for improving the thermal efficiency of a gas turbine? Explain any one method. (5)
- VI. (a) Draw the block diagram of a steam power plant and explain its working. (5)
- (b) What are the various components of a nuclear reactor? Explain the function of each. (5)
- OR
- VII. (a) Flue gases passing through a chimney of a steam power plant is 300°C. Calculate the mass of flue gas when the draught produced is equal to 2 m of water, when the flue gas formed is 23 kg per kg of fuel burnt. The diameter of the chimney is 2 meters. (5)
- (b) Explain the different cooling methods used in nuclear reactors. (5)
- VIII. (a) Derive Betz equation to calculate the theoretical efficiency of wind turbines. What are the various types of losses that decrease the efficiency of wind turbine? What is the approximate net efficiency of a wind turbine? (5)
- (b) Explain the working of two types of solar collectors with neat sketches. (5)
- OR
- IX. (a) Describe the working of an open cycle MHD system with a sketch. (5)
- (b) Explain the working of a geothermal power plant. (5)
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Power Plant Engineering

Time 105 Mins
50

Internal Test -1 -Odd Nos

Max Marks:

(Answer all questions)

- 1 a Define the terms maximum demand, demand factor, load factor, plant use factor & diversity factor .Explain each with an example 5x2
=10
- 2 A generating station has the following daily load cycle :
- | | | | | | | |
|--------------|-----|------|-------|-------|-------|-------|
| Time (hours) | 0—6 | 6—10 | 10—12 | 12—16 | 16—20 | 20—24 |
| Load (MW) | 20 | 25 | 30 | 25 | 35 | 20 |
- Draw the load curve and find
(i) maximum demand, (ii) units generated per day, (iii) average load,
(iv) load factor

OR

A central station is supplying energy to a community through two substations. Each substation feeds four feeders. The maximum daily recorded demands are :

POWER STATION..... 12,000 KW

Substation A 6000 kW

Feeder 1 1700 kW

Feeder 2 1800 kW

Feeder 3 2800 kW

Feeder 4 600 kW

Sub-station B 9000 kW

Feeder 1 2820 kW

Feeder 2 1500 kW

Feeder 3 4000 kW

Feeder 4 2900 kW

Calculate the diversity factor between (i) substations (ii) feeders on substation A and (iii) feeders on sub-station B (15 Marks)

- 2 A generating station is to supply four regions of load whose peak loads are 10 MW, 5 MW, 8 MW and 7 MW. The diversity factor at the station is 1.5 and the average annual load factor is 60%. Calculate :
- (i) the maximum demand on the station.
(ii) annual energy supplied by the station
(iii) Suggest the installed capacity and the number of units

OR

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B. Tech. Degree V Semester Examination November 2014

ME 1506 POWER PLANT ENGINEERING

(2012 Scheme)

Time: 3 Hours

Maximum Marks: 100

PART A

(Answer *ALL* questions)

(8 x 5 = 40)

- I. (a) Define depreciation of power plants and explain any method for calculating annual depreciation.
 (b) Write notes on power house equipments.
 (c) Illustrate the advantages of hydroelectric power plants.
 (d) List out the advantages of gas turbine over steam turbine.
 (e) Derive an expression to find chimney diameter at any cross section of the chimney.
 (f) List the safety measures for nuclear power plants.
 (g) Explain 'solar thermal collection'.
 (h) Describe H_2O_2 fuel cells, with neat sketch.

PART B

(4 x 15 = 60)

- II. (a) A power station is to supply three region of load whose peak loads are 20MW, 15MW and 25 MW. The annual load factor is 50% and the diversity factor of the load at station is 1.5. Determine the following: (7)
 (i) Maximum demand on the station
 (ii) Installed capacity suggesting number of unit
 (iii) Annual energy supplied.
 (b) A base load power station and standby power station share a common load as follows: (8)

Base load station annual output	=	$150 \times 10^6 \text{ kWh}$
Base load station capacity	=	35MW
Maximum demand on base load station	=	30MW
Standby station capacity	=	18MW
Standby station annual output	=	$14 \times 10^6 \text{ kWh}$
Maximum demand (peak load) on standby station	=	15MW

Determine the following parameters for both power stations.

- (i) Load factor (ii) Capacity factor (Plant factor)

OR

- III. (a) Describe four types of tariffs for electricity. (8)
 (b) With a neat sketch, make a note on load curve. Describe different types of load curves. (7)

(P.T.O.)

- IV. Explain different methods for improving thermal efficiency of open cycle gas turbine power plant, with neat sketches. (15)

OR

- V. (a) Explain the layout of diesel engine power plant, with a neat sketch. (7)
 (b) Explain the functions of following systems in a diesel engine power plant
 (i) Fuel system (4)
 (ii) Cooling system (2)
 (iii) Lubricating system (2)

- VI. (a) Describe the fluidized bed combustion system, with neat sketch. (7)
 (b) Explain dust and ash handling system, with neat sketch. (8)

OR

- VII. (a) A steam power plant uses coal 5000kg/hr. The heat conversion efficiency is 30% and calorific value of coal is 7000 k.cal/kg. Calculate the electric energy produced per day. (1kWh=869 k.cal). (7)
 (b) Explain BWR and PWR in a nuclear power plant, with neat sketches. (8)

- VIII. (a) Distinguish between thermoelectric conversion and thermionic conversion system. (8)
 (b) Describe Ocean Thermal Energy Conversion (OTEC). (7)

OR

- IX. (a) Explain MHD power generation, with neat sketches. (7)
 (b) Explain different types of solar collectors, with neat sketches. (8)

- III. A proposed plant has the following daily load cycle: (15)

Time in Hours	6-8	8-11	11-16	16-19	19-22	22-24	24-6
Load in MW	15	45	45	25	60	30	20

Draw the load curve and select suitable generating units from the following available, such as 8000 kW, 16000 kW, 20000 kW and 24000 kW. Up on selecting the generating units, calculate (i) Load factor and (ii) Plant use factor.

- IV. Explain with neat sketches diesel engine power plant layout, and its components. (15)

OR

- V. Explain the different types of combustion chamber design in gas turbine power plants. (15)

- VI. Explain with sketches the different types of coal handling systems employed in steam power plants. (15)

OR

- VII. How are the different types of nuclear reactor classified? Write briefly on each classification. (15)

- VIII. (a) From Bets law find out the maximum percentage of power that can be extracted from wind. What are the other losses and the percentage of such power losses? (10)

- (b) Define solar constant. What is the value of solar constant for earth? (5)

OR

- IX. (a) Explain the principle of MHD power generation. (8)

- (b) Explain the principle of OTEC system. (7)

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B.Tech. Degree V Semester Examination November 2015

ME 1506 POWER PLANT ENGINEERING (2012 scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) What is the significance of incremental rate for a power plant?
 (b) Explain briefly load curve and load duration curve.
 (c) State the advantages of gas turbine power plant over diesel power plant.
 (d) What factors should be considered while selecting a site for diesel power plant?
 (e) What are the advantages of FBC system over the conventional one?
 (f) What are the principal reasons for operating steam turbines with condenser?
 (g) Give classifications of a tidal power plant and list out its components.
 (h) Write a short note on "Wind Electricity Economics".

PART B

(4 × 15 = 60)

- II. Determine the following from the daily load curve as per the table below if (3 × 5 = 15) the max. demand of a power station is 96000KW.

Time hours	0-6	6-8	8-12	12-14	14-18	18-22	22-24
Load (MW)	48	60	72	60	84	96	48

- (i) Determine the load factor of power station.
 (ii) What is the load factor of standby equipment rated at 30 MW that takes up all load in excess of 72 MW?
 (iii) Calculate its use factor.

OR

- III. (a) Explain briefly various methods used to calculate the depreciation cost. (9)
 (b) How can power generation cost be reduced? (6)
- IV. Explain the essential components of a diesel power plant with sketches. (15)

OR

(P.T.O.)

- V. (a) What are the major fields of application and limitations of gas turbine power plants? (9)
- (b) Briefly explain closed cycle gas turbine power plant. (6)
- VI. (a) Discuss suitability of steam plant to supply load in an interconnected system. (8)
- (b) Why are multiple chimneys preferred to single unit chimneys? (7)
- OR**
- VII. (a) List out the advantages and disadvantages of a nuclear plant over conventional thermal plants. (7)
- (b) What are the main purpose of super heater? What are the advantages of superheated steam? (8)
- VIII. (a) Describe working of a solar power plant by using a neat sketch. (8)
- (b) Explain working of a fuel cell and list its advantages over other non conventional systems of power generation. (7)
- OR**
- IX. (a) Describe the working of a geothermal power plant. (8)
- (b) Discuss the merits and limitations of a wind mill power generation unit. (7)

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***B.Tech. Degree V Semester Special Supplementary Examination
August 2015***

ME 1506 POWER PLANT ENGINEERING
(2012 scheme)

Time : 3 Hours

Maximum Marks : 100

PART A
(Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Define the terms: (i) Maximum demand (ii) Demand factor (iii) Load factor (iv) Plant use factor (v) Diversity factor.
- (b) Explain the functions of penstocks draft tubes and surge tanks with figures.
- (c) What are the advantages and disadvantages of diesel engine power plants?
- (d) Draw a two stage gas turbine cycle on a temperature entropy diagram and explain various processes.
- (e) What are the mountings and accessories in a power plant boiler?
- (f) Explain fluidized bed combustion.
- (g) Explain the salient features of flat plate solar collector and evacuated tube solar collector with diagrams.
- (h) Write a short note on fuel cells.

PART B

(4 × 15 = 60)

- II. The daily demands of three consumers are given below.

(15)

Time	Consumer 1	Consumer 2	Consumer 3
Midnight to 8 AM	No Load	200 W	No Load
8 AM to 2 PM	600 W	No Load	200 W
2 PM to 4 PM	200 W	1000 W	1200 W
4 PM to 10 PM	800 W	No Load	No Load
10 PM to Midnight	No Load	200 W	200 W

Plot the load curve and find (i) Maximum demand of individual consumers (ii) Load factor of individual consumers (iii) Diversity factor (iv) Load factor of the station.

OR

(P.T.O.)

B.Tech. Degree V Semester Special Supplementary Examination July 2016

ME 1506 POWER PLANT ENGINEERING (2012 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer ALL questions)

(8 × 5 = 40)

- I. (a) What is the relevance of local factor on different types of consumers?
- (b) What are the suitable site conditions for a Buttress Dam?
- (c) What are the essential components of diesel power plants?
- (d) List out the advantages of closed cycle gas turbine plants.
- (e) What are the different methods for governing of a steam power plant?
- (f) Describe on environmental aspects of nuclear power plants.
- (g) Enumerate three main ways to exploit solar energy.
- (h) What are the advantages of fuel cell over other non-conventional power systems?



PART B

(4 × 15 = 60)

- II. (a) Describe the difficulties to meet peak load. (5)
- (b) A small plant capital cost is ₹50,000/- and its salvage value is ₹5000/- at the end of a useful life of 20 years. Find its value half way through its life based on: (10)
 - (i) Straight line depreciation method.
 - (ii) Sinking fund depreciation at 5% compounded annually.
- OR**
- III. (a) What are the various costs in economics of power plants? (5)
- (b) Describe various types of spillways and conditions applicable for each one. (10)
- IV. Answer the following: (15)
 - (i) Why is supercharging necessary in diesel power plants?
 - (ii) List out the classification of gas turbine plants according to different criteria.
 - (iii) Brief out the working of the governing system of gas turbine power plants.
- OR**
- V. (a) What are the advantages and disadvantages of diesel plants over thermal plants? (10)
- (b) What are the factors considered for selection of underground fuel storage? (5)
- VI. (a) What are fission and fusion processes? Compare them. (8)
- (b) What are the major advantage of FBC system over the conventional one. (7)
- OR**
- VII. (a) Compare fire tube and water tube boilers. (8)
- (b) Differentiate travelling gate stoker firing and underfeed stoker firing. (7)
- VIII. Write notes on the following. (15)
 - (i) Batch type biogas plant.
 - (ii) What are the advantages of fuel cell power generation?
 - (iii) Factors affecting the suitability of the site for tidal power plant.
- OR**
- IX. Write notes on the following: (15)
 - (i) Working of Thermionic conversion system.
 - (ii) What are the disadvantages of wind power?
 - (iii) Different types of concentrating type collectors.